

GabayPoz Recommender System — Version 1.1

Demonstration of the Program-First Pipeline

GabayPoz Project — Team 4 (Model Development)

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Abstract

This document is an executable demonstration of the GabayPoz recommender, version 1.1. It is generated by running `demo.py` against bundled questionnaire/program fixtures under `data/sample/` plus the v1.1 handoff tables under `data/processed/team4_model/`. The script emits a LaTeX fragment (`results.tex`) that this wrapper includes verbatim. Each subsection describes one synthetic student profile, the top-three program recommendations, the primary school selected for each program, rank-1 explanation fields, and the three rows shaped for `model_recommendation`. Rebuilding the PDF reproduces this content from source; see `Makefile`.

1. About this demonstration

GabayPoz is a constraint-aware, explainable college program recommender for Grade 11/12 students in Pozorrubio, Pangasinan. Two rounds of EDA established that the binding constraints on tertiary access are not academic preparation but *geography* (mean barangay-to-university road distance: 76.9 km / 79.2 min) and *household economics* (a $12.7\times$ education-spending gradient across FIES income deciles). The recommender honours those findings by scoring programs first, then selecting feasible schools under each program using barangay-specific travel and affordability tables.

The pipeline runs in six phases:

1. **Student profile.** Q1–Q9 build a six-dimensional affinity vector over (STEM, HEALTH, ARTS, BUSINESS, EDUCATION, AGRICULTURE). Internal-interest questions carry 55% of the student score and aptitude questions carry 45%.
2. **Program fit.** Programs are ranked before school selection using 0.70 cosine similarity and 0.30 normalized dot product between the student vector and program affinity vector.
3. **Market context.** Municipality field saturation is blended as a small 10% context signal. It describes local field presence, not guaranteed job demand; missing saturation data falls back to a neutral score with a warning.
4. **Q12 adjustment.** If the program duration or board-exam burden exceeds the student's tolerance, the program score receives the documented 0.85 soft penalty.
5. **School selection.** For each candidate program, the recommender joins `university_program` to `university`, then applies Q11 commute and Q10 affordability as hard filters to school suggestions.
6. **Output contract.** The recommender returns explanation objects for the UI/API and writes exactly three `model_recommendation`-shaped rows through the caller's write callback when persistence is requested.

The four profiles below exercise the v1.1 contract: a health-oriented student tolerant of longer programs (D1), an arts-oriented student with nearby-only travel (D2), an agriculture-oriented student

with moderate budget and commute tolerance (D3), and a STEM-oriented student with a tight budget and nearby-only travel (D4).

2. Demonstrative profiles

D1: Health-oriented, ready for board-exam programs

Barangay id: 1; Q10: C; Q11: C; Q12: A.

Model id: `tds_recommender_v1_1`; warnings: none.

Top recommendations.

Rank	Program	Primary school	Score	Distance	Commute
1	BS Nursing	Pangasinan State University - Urdaneta City	0.750	14.3 km	13.0 min
2	BS Medical Technology	Pangasinan State University - Urdaneta City	0.672	14.3 km	13.0 min
3	Bachelor of Elementary Education	University of Eastern Pangasinan	0.200	7.3 km	7.8 min

Primary and alternate school suggestions.

- BS Nursing: **Primary** – Pangasinan State University - Urdaneta City (14.3 km, 13.0 min, PHP 22,888). **Alternates** – University of Eastern Pangasinan (7.3 km, 7.8 min, PHP 31,579); Urdaneta City University (16.7 km, 15.6 min, PHP 40,012).
- BS Medical Technology: **Primary** – Pangasinan State University - Urdaneta City (14.3 km, 13.0 min, PHP 22,888). **Alternates** – University of Eastern Pangasinan (7.3 km, 7.8 min, PHP 31,579).
- Bachelor of Elementary Education: **Primary** – University of Eastern Pangasinan (7.3 km, 7.8 min, PHP 31,579). **Alternates** – Urdaneta City University (16.7 km, 15.6 min, PHP 40,012).

Rank-1 structured details.

- Matched dimensions: Health.
- Alternate feasible schools: 2.
- Q10 tier: 5; Q11 commute ceiling: none; Q12 response: A.
- Market context: Health score 0.535618 via `ecosystem_saturation_v1_1`.
- Penalties: –.
- Scholarships at primary school: 2 (CHED Full Merit Scholarship, DOH Nursing Scholarship).
- Low confidence flag: true; reason: LOW_SIGNAL.

Rank-1 explanation. BS Nursing matches your strongest Health signals. Pangasinan State University - Urdaneta City is the primary school suggestion because it passed your travel and affordability limits. Local Health field presence is included only as small context, not a job guarantee.

Persisted rows.

Rank	Program id	University id	Model score
1	2	3	0.750
2	8	3	0.672
3	5	1	0.200

D2: Arts-oriented, nearby-only travel

Barangay id: 1; Q10: B; Q11: A; Q12: B.

Model id: tds_recommender_v1_1; warnings: none.

Top recommendations.

Rank	Program	Primary school	Score	Distance	Commute
1	AB Communication	Pangasinan State University - Urdaneta City	0.762	14.3 km	13.0 min
2	BS Business Administration	University of Eastern Pangasinan	0.328	7.3 km	7.8 min
3	BS Computer Science	Pangasinan State University - Urdaneta City	0.196	14.3 km	13.0 min

Primary and alternate school suggestions.

- AB Communication: **Primary** – Pangasinan State University - Urdaneta City (14.3 km, 13.0 min, PHP 22,888). **Alternates** – University of Eastern Pangasinan (7.3 km, 7.8 min, PHP 31,579); Urdaneta City University (16.7 km, 15.6 min, PHP 40,012).
- BS Business Administration: **Primary** – University of Eastern Pangasinan (7.3 km, 7.8 min, PHP 31,579). **Alternates** – Urdaneta City University (16.7 km, 15.6 min, PHP 40,012).
- BS Computer Science: **Primary** – Pangasinan State University - Urdaneta City (14.3 km, 13.0 min, PHP 22,888). **Alternates** – University of Eastern Pangasinan (7.3 km, 7.8 min, PHP 31,579); Urdaneta City University (16.7 km, 15.6 min, PHP 40,012).

Rank-1 structured details.

- Matched dimensions: Arts.
- Alternate feasible schools: 2.
- Q10 tier: 3; Q11 commute ceiling: 45.0 min; Q12 response: B.
- Market context: Arts score 0.516332 via ecosystem_saturation_v1_1.
- Penalties: –.
- Scholarships at primary school: 1 (NCCA Arts and Culture Grant).
- Low confidence flag: true; reason: LOW_SIGNAL.

Rank-1 explanation. AB Communication matches your strongest Arts signals. Pangasinan State University - Urdaneta City is the primary school suggestion because it passed your travel and affordability limits. Local Arts field presence is included only as small context, not a job guarantee.

Persisted rows.

Rank	Program id	University id	Model score
1	3	3	0.762
2	4	1	0.328
3	1	3	0.196

D3: Agriculture-oriented, moderate budget

Barangay id: 3; Q10: B; Q11: B; Q12: B.

Model id: tds_recommender_v1_1; warnings: none.

Top recommendations.

Rank	Program	Primary school	Score	Distance	Commute
1	BS Agriculture	Pangasinan State University - Urdaneta City	0.798	24.6 km	28.3 min
2	BS Civil Engineering	Pangasinan State University - Urdaneta City	0.169	24.6 km	28.3 min
3	BS Medical Technology	Pangasinan State University - Urdaneta City	0.162	24.6 km	28.3 min

Primary and alternate school suggestions.

- BS Agriculture: **Primary** – Pangasinan State University - Urdaneta City (24.6 km, 28.3 min, PHP 32,140). **Alternates** – University of Eastern Pangasinan (17.6 km, 24.7 min, PHP 40,804).
- BS Civil Engineering: **Primary** – Pangasinan State University - Urdaneta City (24.6 km, 28.3 min, PHP 32,140). **Alternates** – University of Eastern Pangasinan (17.6 km, 24.7 min, PHP 40,804).
- BS Medical Technology: **Primary** – Pangasinan State University - Urdaneta City (24.6 km, 28.3 min, PHP 32,140). **Alternates** – University of Eastern Pangasinan (17.6 km, 24.7 min, PHP 40,804).

Rank-1 structured details.

- Matched dimensions: Agriculture.
- Alternate feasible schools: 1.
- Q10 tier: 3; Q11 commute ceiling: 90.0 min; Q12 response: B.
- Market context: Agriculture score 0.233866 via `ecosystem_saturation_v1_1`.
- Penalties: –.
- Scholarships at primary school: 2 (DA-ATI Agricultural Scholarship, DOST-PCAARRD Agriculture Grant).
- Low confidence flag: `true`; reason: `LOW_SIGNAL`.

Rank-1 explanation. BS Agriculture matches your strongest Agriculture signals. Pangasinan State University - Urdaneta City is the primary school suggestion because it passed your travel and affordability limits. Local Agriculture field presence is included only as small context, not a job guarantee.

Persisted rows.

Rank	Program id	University id	Model score
1	6	3	0.798
2	7	3	0.169
3	8	3	0.162

D4: STEM-oriented, tight budget

Barangay id: 2; Q10: A; Q11: A; Q12: B.

Model id: `tds_recommender_v1_1`; warnings: `none`.

Top recommendations.

Rank	Program	Primary school	Score	Distance	Commute
1	BS Computer Science	Pangasinan State University - Urdaneta City	0.775	18.5 km	19.5 min
2	BS Civil Engineering	Pangasinan State University - Urdaneta City	0.672	18.5 km	19.5 min
3	BS Medical Technology	Pangasinan State University - Urdaneta City	0.396	18.5 km	19.5 min

Primary and alternate school suggestions.

- BS Computer Science: **Primary** – Pangasinan State University - Urdaneta City (18.5 km, 19.5 min, PHP 26,659). **Alternates** – none after Q10/Q11 filters.
- BS Civil Engineering: **Primary** – Pangasinan State University - Urdaneta City (18.5 km, 19.5 min, PHP 26,659). **Alternates** – none after Q10/Q11 filters.
- BS Medical Technology: **Primary** – Pangasinan State University - Urdaneta City (18.5 km, 19.5 min, PHP 26,659). **Alternates** – none after Q10/Q11 filters.

Rank-1 structured details.

- Matched dimensions: STEM.
- Alternate feasible schools: 0.
- Q10 tier: 1; Q11 commute ceiling: 45.0 min; Q12 response: B.
- Market context: STEM score 0.506331 via `ecosystem_saturation_v1_1`.
- Penalties: –.
- Scholarships at primary school: 2 (DOST-SEI S&T Undergraduate Scholarship, CHED Merit Scholarship).
- Low confidence flag: `true`; reason: `LOW_SIGNAL`.

Rank-1 explanation. BS Computer Science matches your strongest STEM signals. Pangasinan State University - Urdaneta City is the primary school suggestion because it passed your travel and affordability limits. Local STEM field presence is included only as small context, not a job guarantee.

	Rank	Program id	University id	Model score
Persisted rows.	1	1	3	0.775
	2	7	3	0.672
	3	8	3	0.396

3. How to reproduce this PDF

```
cd demo
make                # runs demo.py then pdflatex twice
```

`make clean` removes the LaTeX intermediates and the generated `results.tex`; `make distclean` additionally removes `demo.pdf`. Re-running from scratch reproduces the PDF deterministically: the recommender is fully stateless, the sample fixtures and processed v1.1 CSV mirrors are version-controlled, and the ranking rules are documented in `docs/tds.md`.

4. References

- **TDS:** `docs/tds.md` — the authoritative specification this implementation traces to.
- **EDA v1:** `docs/eda_v1.pdf` — Pangasinan education, employment, and access profile.
- **EDA v2:** `docs/eda_v2.pdf` — Pozorrubio program access, scholarship analysis, and employment signals.

- Burke, R. (2002). Hybrid recommender systems. *User Modeling and User-Adapted Interaction*, 12(4).
- Liu, T.-Y. (2009). Learning to rank for information retrieval. *Foundations and Trends in IR*, 3(3).
- Philippine Institute for Development Studies (2019). *Tracer study of CHED-scholarship and non-scholarship graduates* (Discussion Paper No. 2019-26).